

Motivational Aspects in Energy Feedback Systems Design

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ABSTRACT

In harmony with the 3rd HCI wave, the design of residential energy feedback systems (REFS) is an emerging research area in HCI, stimulated by the changes the world has suffered for economic and environmental reasons. Theoretical background supports the ambitious goal of changing a culture by means of a new technology, evidencing the importance of addressing social aspects, beyond those traditionally related to technology. While most concerns are related to sustainable attitudes, fine-tuning the effectiveness of REFS solutions, the scenario addressed by this research requires a deeper analysis of culture to understand particular issues as fraud and the growing of local economy for the design of a feedback system. This paper analyzes technical, formal and informal aspects of this technology introduction in a developing society, inspiring a set of design guidelines and requirements for REFS.

Keywords

Motivation, eco-feedback technology, residential energy feedback systems

INTRODUCTION

Whether for environmental, technical or economical reasons, the world is putting efforts in rapidly reconsidering how to supply, distribute and consume electric power. This new scenario, frequently associated with the concept of Smart Grid [1], may lead to new cost rate structure, for example, stating different rates per kWh according to time of the day, consequently changing habits and people's relation with electricity usage and control. Moreover, it usually brings to households a new technical artifact for measuring and displaying information about energy consumption per periods of time, estimated monthly consumption, and energy costs associated with appliances usage, among other features [2]. Mobile phones, Web, tablets and interactive TV may also be used as indirect displays for showing consumption information, leading to consequences to the HCI domain, especially in its 3rd paradigm, that considers technology as part of our life in

almost all aspects of it [3].

Designing interactive systems to promote sustainable consumption behavior has recently emerged as a key and rapidly growing area of interest within HCI community. Froehlich et al [4] draw upon other authors to define *eco-feedback technology*, as technology that provides feedback on individual or group behaviors with a goal of reducing environmental impact in a variety of domains including energy consumption. Eco-feedback technology is based on the working hypothesis that most people lack awareness and understanding about the way their everyday behavior (e.g. using automatic clothes dryer instead of air-drying their clothes) affect the environment. They argue that an “environmental literacy gap” may be bridged by automatically sensing these activities and feeding related information back through computerized means (e.g., mobile phones, ambient displays, or online visualizations).

Pierce et al [5] argue that, although energy savings of 7-20% have been attributed to the use of *residential energy feedback systems* (REFS), little is known about what specific conservation behaviors do or do not result in such reported savings, how individuals engage or do not engage with feedback, or why conservation does or does not occur in relation to various types of feedback or their design issues. As He et al [6] state, feedback informs, but not necessarily motivates a sustainable behavior.

Field studies evidenced that beyond values and intentions, electricity consumption behavior is embedded in habits and routines, which are patterns of unconscious actions guided by the environment [8,9]. As such, what would be the motivations for using or not such artifact in the context of a developing country, with diverse range of environmental awareness, technology literacy, and societal conditions?

Mentioning Bodker's work [3], Pierce et al [8] suggest emphasizing design' for non-work, non-purposeful, and non-rational uses of technology in the domestic context, according to the 3rd HCI paradigm; moreover they recommend to consider experiential, social and cultural aspects to create an awareness of energy usage, conservation and its relevance for well-being. Dillahunt and Mankoff [7] agree about the importance of addressing socioeconomic factors in design and evidence the importance of considering multiple stakeholders when creating REFS that intend to reach significant audience.

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Considering an anthropological view, Hall [10] states that a technically introduced change in a culture opens the way to form the bases for a new cultural system.

Thus, this research aims at understanding the process of transforming the energy consumption by means of a technology introduction in a Brazilian scenario. More specifically we are considering the context of Minas Gerais and Rio de Janeiro states, which deal with a diversity of cultural aspects, and with issues such as nonpayment and fraud (illegal electricity connection) besides conservation for environmental reasons – motivation usually addressed in literature [4].

In accordance with [6] and [7], which argue that feedback technology design should consider different motivations for saving energy, this work explores the hypothesis that the motivation involved in technology adoption varies according to different needs and expectations of users. We analyze the situated scenario with Organizational Semiotics artifacts aiming at the consideration of motivational aspects in the design of REFS, as well as social and cultural particularities that influence the artifact effectiveness.

The smart metering deployment – recording, transmission and billing of electricity consumption – by two electric companies in Brazil has been considered the case study of this research that aims at developing a design framework informed by motivational issues. The analysis we present stands for a step towards this development.

This paper is organized as follows: the next section conceptualizes challenges addressed in this research such as motivation and culture, and their relation with design; then we analyze the research scenario by applying Organizational Semiotics tools [12]; the next section discusses strategies and requirements for REFS design for this scenario, providing a set of design guidelines that may motivate different users profiles; it also points out further work and the last section concludes the paper.

MOTIVATION, CULTURE AND DESIGN ISSUES

To make the introduction of this technology effective, personal motivations, cultural values and social issues must be addressed to be reflected in design. The introduction of the new artifact may represent an opportunity to change the relationship between people and the energy consumption, addressing also cultural peculiarities of this research scenario such as nonpayment and fraud.

Motivation

Motivation is as an internal force that gives behavior energy and direction [11]. According to theories from Psychology [11], both external reasons, such as money, and intrinsic motives that involves physiological, social and psychological needs may be sources of motivation to achieve the well-being. In the energy consumption scenario, a completely externally motivated person possibly preserve energy to save money; a somewhat externally regulated: “because I should, I am going to feel good about myself”; a somewhat internally regulated: “because it’s important for the environment” and, finally, a personally

committed, or autonomous with internalized motive: “because it reflects and expresses who I am and my values”.

The more an internal motive reflects a person’s values, the more motivated the person is, and the degree of autonomy in behavior determines how motivated a person is by an extrinsic reason. To internalize a value or a behavior, the person needs to make sense of the value or the merit of acting.

User’s (lack of) motivation

As noticed by [2], monetary reasons usually are the main motive for people to save energy and adopt the technology. However, this motivation fades away over time. The theoretical view about motivation [11] leads us to argue that saving money cannot be considered a motivation enough for inducing a deep change in behavior, unless the saved money contribute to well-being.

Thus, identifying and considering people’s most internal motives in the design may stimulate a positive attitude towards the REFS introduction.

Ethnographic interview and contextual inquiry approaches were combined to analyze people’s everyday interactions with energy-consuming products in the USA [8]. The study evidenced how challenging is to change user’s behaviors once daily interactions with energy-consuming devices can be characterized as unconscious – “rules of thumb” – rather than the result of rational decision making. The REFS people received was not enough to change their interest on energy consumptions and values: none of the respondents knew the cost of 1 kilowatt-hour of electricity or their cost rate structure; although ~80% of respondents personally paid their monthly bills, only ~25% were sure about how much they paid. In the same way, people were uncertain about individual devices consumption.

That study also revealed that minor inconveniences and discomforts, such as setting a different temperature for refrigerators or frequently turning off devices when they are not in use prevent people of changing behavior regarding saving energy, pointing out a lack of strong motivations to conserve. Even when this motivation exists, it does not imply people will actually save energy, especially when acting against their well-being [8].

Dealing with culture

According to Hall [10], cultural aspects embrace people’s assumptions about state, economic systems, and so on, and they guide behavior in persistent ways, many of which outside of awareness or beyond conscious control.

People’s culture comprises three different modes – informal, formal and technical – that alternate in dominance and influence learning, awareness, affect, and attitudes toward changes. Technical aspects support formal behavior patterns, around which there are certain informal adaptations. The informal is made up of activities learned in everyday life, which are done automatically. The formal mode is resistant to change from the outside, once it is

regulated by rules. According to Hall, the opportunity to introduce a change in culture is at the technical mode, where changes are easily observed and transmitted to others, establishing basis for a new formal system.

This theoretical background highlights the importance of bringing design issues of the technical artifact into discussion.

Design issues

In order to design a technical device that properly evoke the desired change in behavior, motivational and cultural aspects should be taken into account. The following set of questions synthesizes these design concerns and drives this investigation:

- How to create a sense of awareness of energy usage by means of the device?
- How to identify motivations of different people to adopt (or not) this new technology? How to evoke them with device features?
- Considering the diversity of the Brazilian population, how to propose user interfaces that make sense to as many people as possible?
- How to model and represent aspects of culture reflecting them in the user interface in order to foster a behavior change?
- How to promote a good affective quality in the user's interaction with this new device, aiming at keeping their motivation?

THE RESEARCH SCENARIO ANALYSIS

Grounded in Hall's cultural model [10], Organizational Semiotics (OS) [12] provides artifacts to analyze this research scenario from diverse points of view. Within the

OS frame of reference, a technical solution is involved by a formal layer of information (e.g. rules, regulation), which by itself is involved by an informal layer (e.g. underlying intentions, beliefs). The technical layer impacts on the external layers and is also influenced by them, suggesting that an innovation introduction may fail when considered only at the technical level if it is not compatible with people's expectations, culture or current regulation.

Three OS artifacts were applied to support the analysis, which was based on meetings and interviews with electric companies employees and also on perceptions obtained by observing the customer care services. The call centers, on-line chat and in-person attendance offices were visited to provide an understanding of the relationship between the consumers and the companies from different points of view.

Figure 1 illustrates the first artifact, the Stakeholders Analysis Diagram. Based on the technical, formal and informal layers concept, this OS tool maps the stakeholders of the Smart Metering and REFS introduction according to their roles as interested parties.

Among the stakeholders in the different layers of the "semiotic onion" we can see those who **contribute** more directly with the system under analysis (e.g. the displays user interface designers), those who are **source** of information (e.g. the consumer himself/herself), those who represent the **market** (e.g. autonomous electrician technicians, which are opinion makers about this technology), or the social forces of **community** (e.g. media, that can positively promote this change or not, and the association of energy distributors). Stakeholders identified in italic represent departments of energy companies.

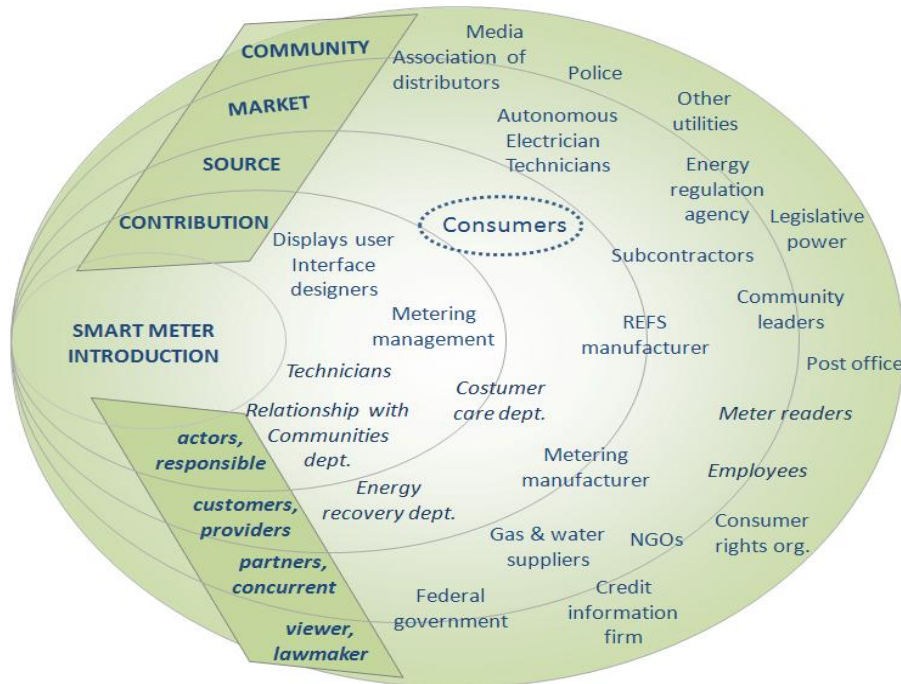


Fig. 1. Stakeholders Analysis Diagram of the research scenario

The stakeholder analysis enables a systemic view of different forces of information regarding the interested parties. In addition to addressing technical, regulatory and social aspects that underlie this technological change, this analysis also considers commercial losses by the power provider mainly due to nonpayment and fraud, in a way to prevent those losses with the introduction of the new technology.

The second OS artifact is the Evaluation Frame [13], represented by Table 1, applied to identify the main concerns of stakeholders, prospecting potential problems the introduction of the smart metering display would cause to different interested parties.

Table 1. Evaluation Frame

Stakeholders	Concerns
Contribution level	
Meter management	- How to identify useful and necessary information to the consumer?
Display user-interface designer	- How to present this information in a way adequate to the diversity of users?
Technicians	- How to arouse user's interest in using the technology?
Relationship with communities	- How to aggregate cultural and motivational values in the technology?
Customer care dept.	- How to inform individual and collective benefits of a conscious consumption?
Source level	
Consumers	- How to stimulate regularization for fraud situations with the technology introduction?
Customer care dept.	- How to motivate a spontaneous usage of the displays by consumers?
Energy recovery dept.	- How to promote environmental and ecological values associated with energy consumption?
Metering and REFs manufacturers	- How do the consumers perceive the new technology? - How to take advantage of different kind of displays (Interactive TV, mobile phones and the Web) for different customers profiles?
Market level	
Autonomous Electrician	- How to involve technicians and installers to promote the technology?
Technicians	- What are other utilities providers (gas and water) interests on this technology? How to address these interests in the display?
Subcontractors	- How to educate and involve employees turning them into supporters?
Metering and REFs manufacturers	- How to encompass community leaders to promote this technology?
Gas & water suppliers	- How to tackle social benefits provided by the government on energy costs on device?
Community leaders	
Employees	
Meter readers	
Federal government	
Association of energy distributors	
Energy regulation agency	

Community level	
Media	- How to announce the benefits and implications of this technology for the customer?
Police	- How to build and keep a positive image of the change in society?
Federal government	- How to address meter readers' concerns about their jobs due to smart metering introduction?
Association of distributors	- How to map consumer protection agencies expectation in the solution?
Energy regulation agency	How can they contribute to the solution?
Community leaders	
Employees	
Meter readers	
Post office	
Consumer right org	
Credit information Firm	

The informal level

To provide ideas or solutions to concerns identified at the Evaluation Frame, both cultural and motivational issues – represented at the informal layer [12] – must be taken into account to represent the way people tend to behavior, their values and beliefs.

In this research scenario, both fraud and nonpayment by the consumers are responsible for a commercial loss to the energy distributors higher than a couple of billion dollars per year [14]. For this reason, these companies are strongly interested in educating people to consume energy more efficiently, in such way that they afford the bill. Also, it was evidenced by [15] that in some areas fraud incidence has not been related to economic aspects, but has been mainly associated to low level of urbanization and violence, somehow related to values acquired at formal education, but strongly disseminated by informal aspects of culture.

Some low urbanized areas that used to be dominated by criminal organizations are now under the police control, streets and informal address have been regularized, bringing forth the sense of citizenship and providing unique opportunity to reestablish the relationship between the energy company and the population.

Within our scenario, the “green consciousness” is not a strong aspect of the culture, mainly due to recent growth of economy, allowing people to access more consumer goods, leading to higher consumptions of energy, fact that affects especially the economically less privileged people [16].

People usually value energy as expensive, having a high cost-benefit relation [17]. The conscious involvement people have with the energy usage usually happens only once a month, when they receive the bill. However, according to the INAF [18], a measure of the functional literacy level among Brazilians aged 15 to 64, 70% of the population has incomplete literacy, which means no skills for reading charts, among other constraints; this may represent also low understanding of concepts applied on energy consumptions such as percentage and kWh.

Informal norms descriptions

As key concepts, we go further into socio-cultural aspects, describing them according to the Norm Analysis, the third OS artifact applied in this study. According to OS [12], an organization is a social system in which people behave by conforming to a system of norms, defined as regularities of perception, behavior, belief and value, that are exhibited as customs, habits, patterns of behavior and other cultural artifacts.

Table 2 presents a formal description of these behavior patterns of informal level in the format of norms [12]: WHENEVER <state> THEN <agent> IS <deontic operator: must, may, must not> TO <action>. This set of norms represents people's current behavior regarding the subject.

Table 2. Descriptions of some informal level norms

	whenever	then	is	to
Always	there is a governmental social benefit	low-privileged consumers	must	not exceed a consumption limit to get a considerable discount on energy bill
	want to buy a new appliance	low-privileged consumers	should	prioritize low-energy consumption appliance
	cannot afford the bill payment	consumers	should	learn how to consume less
Most of times	in debt to the energy company	consumers	may	prioritize to pay other bills instead of paying off the energy debt
		consumers	may	take advantage of the company's criteria to suspend the energy to manage when to pay the bill
	asked about the value of energy	consumers	may	state that energy has a high cost-benefit relation
	lives in a non-urbanized community or a violent area	consumers	may	pay for an (illegal) organization for using an irregular connection to receive energy at home
	surrounded by fraudsters	consumers	may	not evaluate fraud as a crime
	in contact with people who do not pay for energy	consumers	may	believe that they should not pay for energy too
	have incomplete education level	consumers	may	not understand the concepts associated with the energy consumption and its bill (kWh), %, charts
	find a fraud condition	technicians, employers or other consumers	should	denounce for energy company

Frequently	cannot afford the energy bill	consumers	may	adopt an illegal alternative to receive energy at home influenced by local culture
	have a household budget increase	low-privileged consumers	can	buy more new domestic electronic devices
	reflect about the energy provision service	consumers	may	associate the energy supply and distribution with a governmental service
	value governmental services	consumers	may	associate it with tax, misuse of public resources and corruption

Even though it has provided a broad understanding of the scenario, this analysis evidenced that a unique stakeholder representing the consumer does not express different kinds of motivation and relationship with the energy distributor. Based on analysis of customer care services (call center, chat and in-personal assistance), and also interviewing customer attendances, we identified five categories of consumers as represented in Figure 2.

Residential consumers may be regular – up-to-date with the payment of energy bills – or irregular, in fraud condition. In the intermediate state are the debtors, regular customers on nonpayment status, considered as potentially new fraudsters when they cannot control or afford their energy budget.

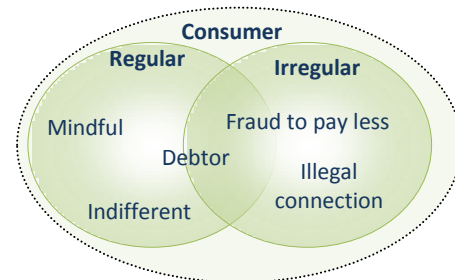


Fig. 2. Consumer profiles

Analyzing consumers' motivations, we identified different types of regular consumers according to the main reason why they control (or not) their energy budget:

- Consumers mindful about energy consumption for environmental reasons (MENV): besides individual interests, this profile wants to conserve natural resources. They act according to their values, representing a strong intrinsic motivation to keep this behavior.
- Mindful about energy consumption for saving money (MMON). To be considered a motivation indeed, the amount of saved money has to be enough to compensate the effort for saving. People who receive governmental social benefits are examples of this profile, once they

must consume below determined target to receive a considerable discount in the energy bill.

- Indifferent (IND): are people who are not concerned with the money they spend on energy bills neither with the natural environment. Usually they believe that the comfort the energy provides is worth the money and do not want to worry about that.
- Debtor (DEB): must be aware of the consumption to leave this condition and afford the bill payment. Usually they want to have no restriction on Credit Information firms to get credit for buying.
- Fraud for paying less (FPL): official customers that make a fraud for paying less than the real consumption. Cultural reasons and impunity make them believe that they would never be caught. In less privileged areas may represent informal businesses, such as popular hairdressers or laundries that would not afford the bill to keep their business. It may also represent influent people that believe in impunity.
- Completely illegal costumer (ILC): both for cultural influence (“everybody does”) or due to the low level of urbanization, sometimes this kind of fraudster is not completely aware of the criminal attitude and just need a good opportunity to be educated and regularize their connection.

For many people, monetary or environmental incentives may not be enough to motivate one to conserve or to adopt a technical artifact [6,9]. Thus, understanding each motivation to keep current behavior, we intend to identify motivations specific of our scenario to change the behavior towards the efficient and regular consumption by adopting a REFS. In Table 3 we describe these motivations and relate them to other stakeholders capable of influencing the success of the technology introduction.

Table 3. Motivations of identified user profiles

	Most likely motivation to keep current behavior	Relationship with stakeholders to motivate a change of behavior towards more efficient and conscious consumption
Mindful	Environmental issues (intrinsic motivation)	May motivate people in the household and people in the society when associated or in the role of Community leaders from associations, schools, and/or NGOs.
	Save money (extrinsic motivation)	The Federal Government and other representative organs, such as regulatory agency, may stimulate consumption reduction by creating policies of benefits. The energy company needs to publish and stimulate this incentive.
Indifferent	Well-being	Features associated to home-automation when aggregated on the REFS may improve comfort and consequently reduce consumption. It must be considered by display manufacturers and all value chain.

Debtor	Lack of information about energy and electronics consumption	Need to be educated by the supplier (by Costumer Care Dept., Technicians – autonomous or not – and media) about how to keep energy consumption in an affordable level. Consumer protection agencies (PROCON, IDEC) should support initiatives to avoid nonpayment.
Fraud pay less	Culture + Subsistence	Employers, Technicians – autonomous or not – the Media and Subcontractors must educate about the consequences of fraud; Police, Law Makers and the society should learn that when everybody pays, the cost of energy could be reduced [14].
Illegal connection	Low level of urbanization	Once living in a recently urbanized area, the customer can receive the regular service, including their bill properly delivered by the Post office. Well hosting new consumers, the relationship with community dept. should promote the advantages of being regular, such as better quality of service and the energy bill as an official proof of address, considered a meaning of citizenship by society.

Beyond orienting stakeholders actions, the identified motivations must be taken into account also in REFS design. To evaluate this impact, we considered strategies and requirements for design extracted from literature to analyze them according to the identified motivational aspects.

STRATEGIES AND REQUIREMENTS FOR REFS DESIGN

When designing a technical device, HCI researchers are traditionally concerned about the understandability, aesthetic, and perceived usefulness of the artifact. The design of REFS brings to light additional challenges such as users’ motivation, attentiveness, cognitive load, information relevancy, and cost [4], as well as their effectiveness on influencing users’ attitude towards the environment.

In the captology domain [19] – field that investigates how digital systems can be applied to change attitudes and behaviors –, a technical device may influence people in three different ways: 1) as a tool that provides information for decision making, increases self-efficacy and changes mental models; 2) as a medium: providing learning and understanding of cause-and-effect relationships, insightful visualizations and sensations; and 3) as social actor establishing social norms and invoking social rules. This set of “*persuasive affordances*” [19] should guide the design of the artifact.

The set of 8 design strategies proposed by [20] addresses these aspects also evidencing that an effective solution is not exclusively related to the REFS features, but to a set of complementary actions on the social domain. These strategies were analyzed with our focus on the user motivation and the previous analysis of our specific scenario:

(1) Offer behavioral cues and indicators of current consumption.

Feedback is a crucial component in learning, which may help all consumer profiles on creating the culture of thinking about energy consumption. The frequency the feedback system is updated is important to ensure credibility – a key aspect that increases the persuasive power of the device [4,19]. Debtors who need to learn how to consume are specially benefitted by this feature.

The granularity of the data (whole house, rooms, each appliance), the accessibility and consequent adequacy with the medium through which information is provided, such as Web, mobile phone, specific device, and so on [4], are also important design factors for consumption feedback.

The aesthetic of the artifact may stimulate its spontaneous usage, representing a possible motivation for the indifferent costumer as well, especially for the young people, who may influence the other family members.

According to [21], iconic and metaphorical images suggest more awareness and motivation for future behavior changes through emotional attachment; although indexical representation is beneficial for informative purpose.

Choosing measurement units to represent consumption (i.e. colorful lights with meaning, currency, kWh, carbon footprint, etc.), as well as the possibility of customizing the display with colors or symbols, for instance, may also promote aspects of autonomy, so that consumers act according to their values and self-identity. The device may lead to a positive feedback when the consumption is lower than expected, and induce positive emotions via initial exposure to technology [22].

Studies have indicated that it is possible to achieve savings in the order of 10% with real-time feedback [8], which is still considered a marginal effect [4].

(2) Provide tools for analysis; provide information about energy use patterns over hours, days, weeks, or months, as well as set target reduction goals to work towards.

To retrospective purpose, indexical representation was considered beneficial [20], which may imply on charts and tables – representations that create barriers for functionally illiterate people (around 70% of Brazilian population). Thus, one of the challenges we will have to face is how to provide information visualization aiming at promoting the user's competence – the design for optimal challenge – in interaction [22,23].

Art and aesthetics have been considered as important concepts for information visualization [20], especially when the purpose is to achieve a good affective quality of user interface [23,24,25]. According to [26], traditional analytic and pragmatic visualization tools have sought to minimize distortions and to foster immediate understanding, while an artistic visualization has the power to keep users interest and also to express a point of view;

this is supported especially when designers are dealing with “*valid reasons to want to change the way people think and it may be that much of the value of visualization comes from its ability to change attitudes*”.

To achieve the ideal level of competence for each user, it is crucial to find the balance, showing charts in a simple way and applying metaphors – i.e., an amount of lamps to represent a number of kWh to help low literate people, for example, but also creating opportunities to analyze consumption and compare it with past behavior [5] for those who are really concerned.

It is worth mentioning that modes of interaction and the information presentation also affect the effectiveness. The way a reduction target (or a baseline) is presented may stimulate users to maintain this level of consumption instead of reducing it [5].

(3) Create social incentive to conserve (i.e., competitions).

A study conducted by [27] resulted in 32% of reduction in electricity usage in a college when a competition was proposed [20]. Social software can also be applied for creating incentive to change behavior. This approach to efficiency represents a new research topic to be explored [4], with an additional challenge in our research scenario to also include low digitally literate people.

Competitiveness in general is a cultural aspect that must be better evaluated in the considered research scenario. It addresses the following “*motivational affordances*” [22,25]: it represents human social bond; facilitates one's desire to influence others; facilitate one's desire to be influenced by others.

(4) Connecting behavior to material impacts of consumption.

This guideline suggests connecting individual actions (e.g. turning off the lights) with the negative consequences of our collective actions, such as consequences for the planet (global warming) [20].

Visual metaphors as well as visual appeal may help on creating a connection between consumption and environmental impacts [20,21].

It is important to investigate the common sense of environmental impact associated do energy consumption in Brazil, once the energy is mostly generated by hydroelectric plant, less polluting than thermoelectric plant, a common power generation worldwide. Probably the connection to the construction of new dams has a higher impact than CO₂ emission, usually applied in most studies.

Creating this connection is a way to develop environmental values, promoting autonomous behavior instead of developing a sense of control, the behavior determined or pressured by others [11,23].

(5) Encouraging playful engagement and exploration with energy.

In [20], the importance of designing for ludic activities, motivated by curiosity, exploration and reflections is highlighted. This guideline is also important as a learning tool, especially when acting in a collaborative way, as stated by [2]. Teachers and parents may use games for influencing others, facilitating human-human interaction [22], and promoting intrinsic motivations associated to relatedness [23]. It may also induce intended emotions via intensive interaction with ICT [22]. Home-automation features may also promote conservation leading to comfort and well-being, such as setting proper light according to the use of an internal environment.

- (6) *Project and cultivate sustainable lifestyles and values: the aesthetic form is something the owner would be proud to display, serving as a symbol of sustainable values and lifestyle to others.*

Both social networks and houses may promote one's identity (and the self [23]) when it is related to sustainable lifestyles. This feature also facilitates one's desire to influence others [22], a typical characteristic of the mindful for environmental reason profile in the role of a community leader.

- (7) *Facilitating discussion and raising public awareness.*

This guideline is related to (4), and suggests that aggregated data of small granularity can be effective in persuading people to conserve. Emphasizing both aggregated economy of a house and showing the collective results helps on building values so that people may act with autonomy due to reducing consumption. Large displays in public places may contribute to the success of a local initiative of reduction.

Figure 3 illustrates an example of a panel in Rio de Janeiro, aiming at reducing the public garbage daily collected in the beach area. The panel displays how many grams of trash were collected per capita per day and the monthly total in tons. Similar initiatives on Web and social networks should also be evaluated.

This guideline also contributes to develop environmental values, promoting autonomous behavior instead of developing a sense of control [23].



Fig. 3. Garbage collected panel in the beach

- (8) *Stimulating critical reflection.*

For [20], critical approaches to REFS design can also lead to more pragmatic behavior and changes, as the example of a REFS that forces users to estimate how much energy is being consumed before displaying the actual usage [28]. This guideline should be associated with games and social networks usage facilitating learning on how to consume and specially how to buy new appliances.

Design guidelines

Based on the design strategies proposed by [20] and in the analysis of this research context, Table 4 presents design requirements, relating them to the consumer profiles that would be motivated by the requirement.

Table 4. Design requirements to motivate users

Requirement	Most benefitted profiles
(1.1) Show the current level of consumptions in an attractive (good affective quality of the device) and customizable way	DEB, IND
(1.2) Show consumption in different ways: monetary value, relation with environment, kWh	MENV, MMON
(1.3) Explore the strength of the device (Web, interactive TV, smartphone) to show consumption information, but aiming at keeping the consistence between the user interfaces	All users
(2.1) Show consumption patterns in accessible and attractive look-and-feel. Make it possible to execute assistive technology when necessary	
(2.3) Stimulate consumption below the target	
(2.4) Alerts of bad debt and over consumption	MMON, DEB, FPL, ILC
(3.1) Explore the association between illegal attitudes and the energy price	
(3.2) Share public commitments trough social software	MENV, DEB, ILC
(4.1) Associate collective consumptions with new dams building needs	
(4.2) Lights on a public place, for instance, such as an art object in a main square or even illuminating the Cristo in Rio, whether in virtual and real world, showing peaks of consumption	MENV, IND, ILC
(5.1) Design Games that stimulate collective playing	MENV, MMON, IND
(5.2) Funny interactions with devices to learn concepts associated to energy generation and distribution	MENV, MMON, DEB
(5.3) Promote well-being with home-automation features that optimize energy usage	IND, FPL
(6.1) Show stamps and awards in social networks or in the houses	MENV
(7.1) Show individual achieved economy	MMON, DEB, ILC
(7.2) Publish global results of individual attitudes (in a community, city, state) whether in real or virtual world	MENV, IND
(8.1) Design Games that stimulate and motivate lower consumption, i.e. about how to choose new electronics to buy	MMON, DEB

Although individual motivations are important aspects to be addressed in REFS design, it cannot be considered as a single user device. Participants of the study conducted by [2] found important to use it in collective way to evoke discussions about the interaction and energy usage, reinforcing the need to fit the device in the household routine.

For this reason, the same device should be featured to motivate different people (such as children and parents) and to keep users' interest in the long term, highlighting the importance of indirect displays such as smartphones and the Web to envisage the differences.

The design guidelines we proposed are mostly represented at the technical level [10], but also pervade the formal and informal levels to promote a solution complemented by social actions. In particular, the social networks may represent an important aspect for an effective solution, due to the relevance of this kind of software to the society.

Froehlich et al [4] argue that most HCI studies are narrowly concerned with the artifact design, while in environmental psychology, investigation is mainly related to the effect of the feedback intervention [29], evidencing a gap between the disciplines in linking achieved impact on behavior to specific design aspects. This work represents the first step towards bridging this gap in the Brazilian context; in the sequence, proof of concepts for different kinds of displays and solutions in different real environments will be explored to analyze the intelligibility and motivational aspects of their user interfaces. Both qualitative experiments in the field and in the laboratory, and quantitative enquiring with real users will be conducted to analyze the potential consequences of this kind of technology introduction. Similarities and distinctions between consumption of electricity and water and gas will also be considered in future works, following the trend to present all utilities in the same device regarding the promotion of more sustainable behavior.

CONCLUSION

Current HCI research on residential energy feedback systems design focuses on the design of displays mainly with sustainable or environmental appeal. This research aimed at identifying different possible user's motivation for changing behavior towards energy consumption to reflect this on the REFS design.

Although we recognize cultural differences among specific regions of our scenario, e.g. fraud is more socially accepted in some places than in others, we intended to build a model that represents a developing society as a whole, making it applicable to other locations.

Five consumer profiles were identified in our specific scenario, reflecting their motivations and relation with the energy company, taking into account social aspects such as fraud and nonpayment.

A set of requirements for the REFS was established to guide the user interface's design and features that may

promote positive social actions. The next steps in this work involve giving shape to REFS prototypes informed by the results reported here.

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